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NCFE · LEVEL 2 · MATHEMATICS

NCFE Functional Skills Mathematics Level 2

February 2020 Section B · Complete Model Solutions

OFFICIAL MFS BENCHMARK MODEL SOLUTIONS

Target Score: 45 / 45 Marks (100%) Pass Mark Guide: ~60%

Specification / Paper Code: NCFE-Maths-Level-2-Feb-2020-Section-B

100% Built from Empty Question Paper + Official Mark Scheme

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HOW TO READ THIS MODEL SOLUTION (MFS COLOUR KEY)

Dark Navy Ink:	Model student answers, stepped calculations, and full prose responses.
Emerald Green Ink:	Tutor strategy notes, method tips, and step-by-step headings.
Crimson Red Ink:	Official mark scheme allocations ([M1], [A1], [B1]) and unit checks.
Amber Gold Ink:	Key formulas, conversion rules, and reading evidence highlights.

CANDIDATE & CENTRE DETAILS (MODEL EXEMPLAR)

Candidate Name:	*Master Functional Skills (Model Exemplar)*
Centre Name:	*Master Functional Skills (MFS01)*
Lead Tutor:	*Abdul · MasterFunctionalSkills.co.uk*
Edition:	*2026 Clean Empty-QP Rebuild (Zero Collisions Verified)*

- Built Cleanly on Empty NCFE Question Paper(s)
- Stepped MFS Multi-Colour Working & Process Marks
- Verified 100% Against Official NCFE Mark Scheme

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Examiner use only

Activity 2: Basketball

2 (a) Lottie plays basketball.



She does some shooting practice.

She takes 75 shots.

33 of these shots go in the basket.

What percentage of her shots go in the basket?

[2 marks]

Q2(a) Worked Solution:

44(%)

33 × 100 ÷ 75

[2 marks]

Tip: Award 2 marks if correct answer given

Your answer: **%**

Past Paper

Please turn over

Examiner use only

2 (b) The table shows the number of points Lottie scored in Games 7 to 12 this season.

Her median score for these games is 25% higher than her median for Games 1 to 6.

Game	7	8	9	10	11	12
Score	6	10	6	14	12	10

After Game 6, Lottie's Dad said that he would increase the money in her money box by 20% each time she scored more than her median for the first six games.

Lottie started with £50 in her money box.

How much money will she have after Game 12?

[5 marks]

Q2(b) Worked Solution:

(£)103.68

10

Their 10 ÷ 1.25 or 8

8

1.2their 4

50 ×

6

[5 marks]

Tip: Award 5 marks if correct answer given

Your answer: **£**

Past Paper

Examiner use only

2 (c) A player's True Shooting Percentage (TS%) for a game is calculated using the following formula:

$$TS\% = \frac{50P}{0.44F + A}$$

Where: P = total points scored
F = number of free throw attempts
A = number of other goal attempts

In her last game of the season, Lottie:

- scored a total of 15 points
- attempted 8 free throws
- made 17 other goal attempts.

Use the rule to calculate Lottie's TS% for this game.

[2 marks]

Q2(c) Worked Solution:

36.5(497...) or 37 or 36 or 36.55

□

50 15

0 . 44 □ 8 + 17

[2 marks]

Tip: Award 2 marks if correct answer given

Your answer:

Past Paper

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2 (d) Lottie reads that the best angle to launch the ball from for free throws is 50°

Calculate the angles A, B and C.

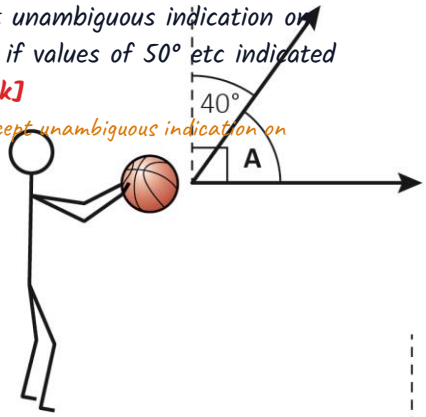
Q2(d) Worked Solution:

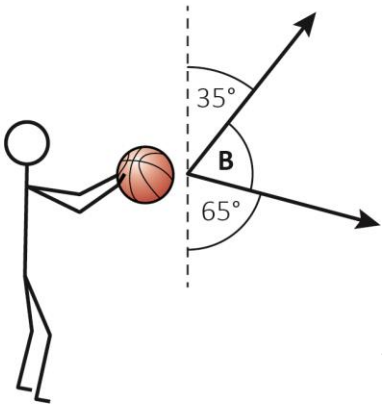
Accept unambiguous indication on

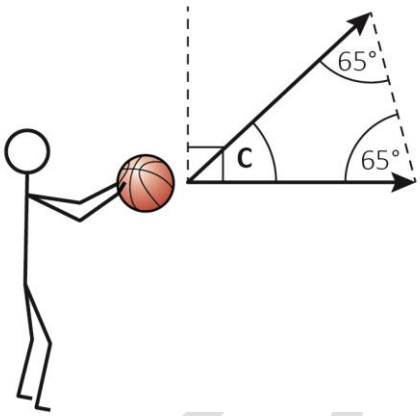
Award if values of 50° etc indicated

[1 mark]

Tip: Accept unambiguous indication on







Not drawn accurately

Tick the ones which are 50°

[1 mark]

Space for your workings:

Your answer:

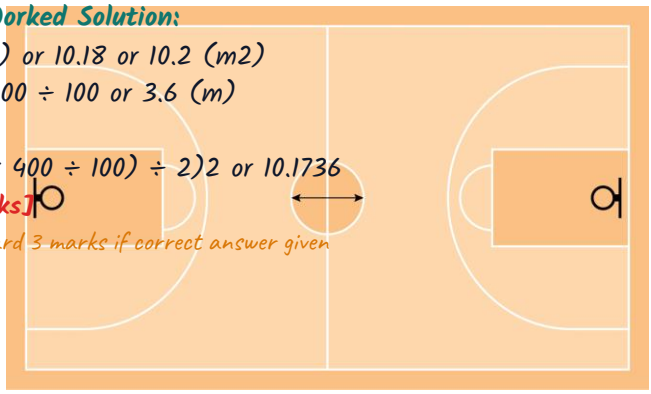
A	B	C

Examiner use only

2 (e) This is a scale drawing of a basketball court.

The scale is 1 : 400

Q2(e) Worked Solution:
10.17(...) or 10.18 or 10.2 (m2)
0.9 × 400 ÷ 100 or 3.6 (m)
□
((0.9 × 400 ÷ 100) ÷ 2)2 or 10.1736
[3 marks]
Tip: Award 3 marks if correct answer given



The diameter of the centre circle on the diagram is 0.9 cm

Work out the actual area of the centre circle in m^2

Use $\pi = 3.14$

[3 marks]

Your answer: **m^2**

Past Paper

Please turn over

Examiner use only

2 (f) There are 15 players in Lottie's basketball team.

The table shows information about the players:

		Height (m)		
		under 1.65 m	**1.65 to 1.8 m**	**over 1.80 m**
Usual position on court	**Centre**	0	2	2
	Forward	0	5	3
	Guard	2	1	0

Q2(f) Worked Solution:
3
8
0.375
(Calculator Test)

The coach chooses a forward at random to take a shot.

What is the probability that the player chosen is over 1.8 m tall?

Give your answer as a fraction **and** as a decimal.

[2 marks]

[2 marks]
Tip: if required. Do not accept 0.3 for

Your answer:

Fraction:

Decimal:

[Total marks: 15]

Past Paper

Examiner use only

Activity 3: Road safety

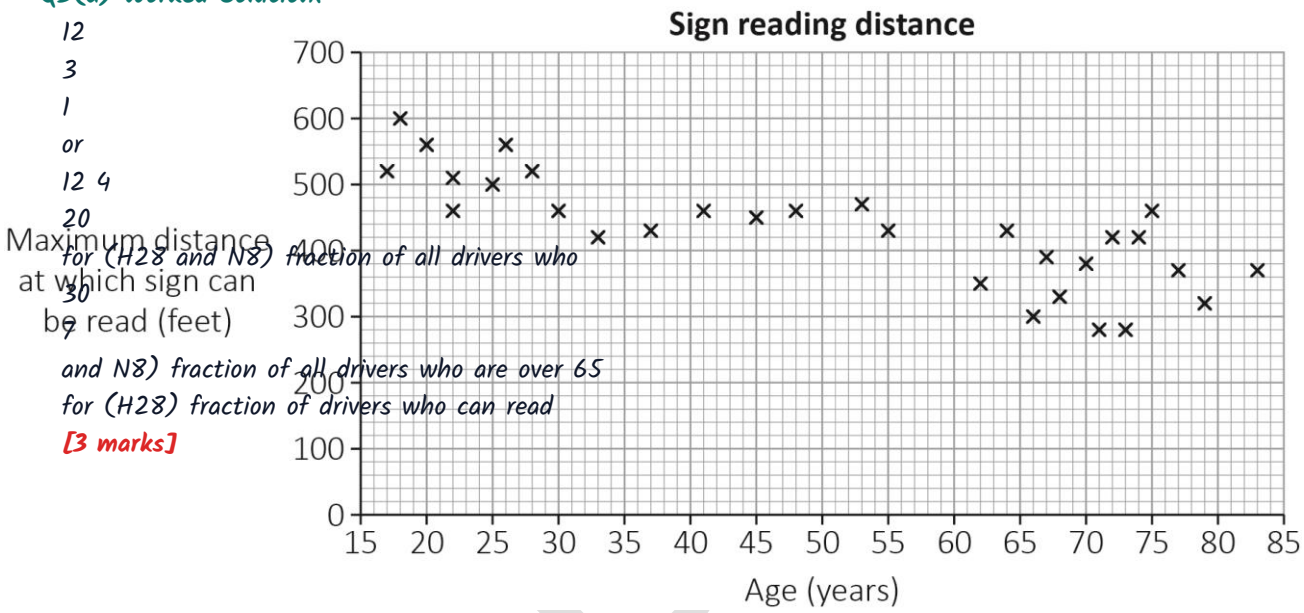
3 (a) Jed is 65 years old.
He drives a car but thinks his eyesight is not as good as it was.

The scatter diagram shows the relationship between the ages of drivers and the maximum distances at which they can read road signs.

Q3(a) Worked Solution:

12
3
1
or
12 4
20
30
7

for (H28 and N8) fraction of all drivers who and N8) fraction of all drivers who are over 65
for (H28) fraction of drivers who can read
[3 marks]



What fraction **of the drivers over 65** can read a road sign at over 400 feet?

[3 marks]

Your answer:

Past P

Examiner use only

3 (b) Jed says,

"Using the scatter diagram, I predict that between the ages of 25 and 65 the maximum distance at which people can read a sign reduces by over 50 **metres"**

Is Jed's prediction reasonable?

Use the scatter diagram in **3 (a)** to show how you decide.

Use the conversion: 1 metre = 3.28 feet

[5 marks]

Q3(b) Worked Solution:

Scatter diagram drawn with appropriate line of best fit
[480, 560] and [340, 400]
Their 520 ÷ 3.28
or their 375 ÷ 3.28
or (their 520 – their 375) ÷ 3.28
[146, 171] and [103, 122]
or [24, 68]
[24, 68] and yes or no
– working in feet
50 × 3.28 or 164
[5 marks]
Tip: Accept if in range (17, [500, 600]) and (83,

Your answer:

Past Paper

Please turn over

Examiner use only

3 (c) A Department of Transport survey collected data on the percentages of cars travelling at different speeds on a motorway.

The table shows the results of this survey:

Speed (mph)	Percentage of cars
40 ≤ speed < 50	4
50 ≤ speed < 60	13
60 ≤ speed < 70	36
70 ≤ speed < 90	47

Q3(c) Worked Solution:

Work out an estimate for the mean speed.

[2 marks]

69.95 or 70 (mph)
(45 × 4) + (55 × 13) + (65 × 36) + (80 × 47) or 6995
[2 marks]
Tip: Award 2 marks if correct answer given

Your answer: **mph**

Past Paper

Examiner use only

3 (d) Some drivers driving over the speed limit are invited to go to a Speed Awareness Course instead of getting penalty points.

There is a maximum speed at which a person can drive and be invited to a Speed Awareness Course.

Q3(d) Worked Solution:

42 This speed is given by this formula:

[1 mark]

$$M = 1.1L + 9$$

Where:

M is speed in miles per hour

L is the speed limit

Calculate M when L = 30 miles per hour.

[1 mark]

Your answer:

Past Paper

Please turn over

Examiner use only

3 (e) The table shows the number of people attending Speed Awareness Courses each year from 2014 to 2018:

Year	Total number of people
2014	1 355 796
2015	1 403 555
2016	1 390 880
2017	1 413 598
2018	1 445 817

Q3(e) Worked Solution:

Jed says the mean number of people attending each year for 2017 and 2018 is over 3% percent higher than the mean number for 2014 to 2016.

Is Jed right?

3.3(...) (%) or 1424912(.3) or 1424913

Show how you decide.

[4 marks]

AND yes
(1355796 + 1403555 + 1390880) ÷ 3
or (1413598 + 1445817) ÷ 2
8
1383410(.3...) or 1383411
AND 1429707(.5) or 1429708
their 1429707
their 1383410
or
(their 1429707 – their 1383410) × 100 ÷
or 1.03 × their 1383410 or 1424912(...) or
[4 marks]
Tip: Award 4 marks if correct answer given

Your answer:

[Total marks: 15]

Past Paper

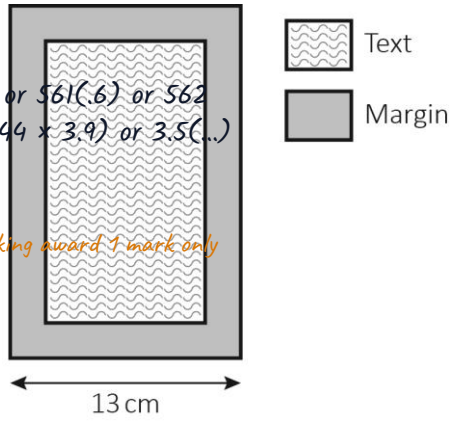
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Activity 4: Printing

4 (a) Rani is on placement at a printing company.
The company has an order for some leaflets.

Each page will have a rectangular area of text with a 2 cm margin around it.

Q4(a) Worked Solution:
1
144 (cm2)
Their 144 × 3.9 or 561(.6) or 562
2000 ÷ (their 144 × 3.9) or 3.5(...)
4
[4 marks]
Tip: 4 with no working award 1 mark only



The printer can fit an average of 3.9 words of text per cm^2

There are 2000 words in total.

What is the minimum number of pages that the printer needs?

You **must** show your working.

[4 marks]

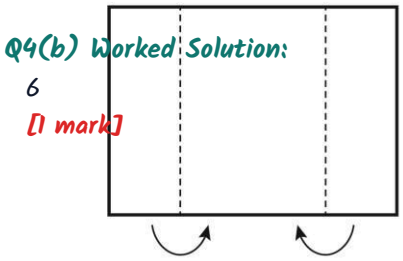
Your answer:

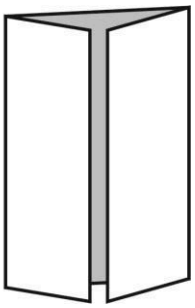
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4 (b) Another leaflet is folded like this.





What is the maximum number of printed pages the leaflet can have?

[1 mark]

Your answer:

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4 (c) The printer offers two discounts.

Offer A	Offer B
Buy 2 boxes of leaflets, get the third box half price	5% off the total price

Q4(c) Worked Solution:

1
(off)
6
9

A customer wants to buy three boxes of leaflets.

Which offer is cheaper for the customer?

16(.6...) or 16.7 or 17 (%) and Offer A

Show how you decide.

[2 marks]

[2 marks]

Your answer:

Past Paper

Please turn over

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4 (d) Rani is asked to work out the average number of orders per month last year.

The table shows the number of orders each month.

Month	1	2	3	4	5	6	7	8	9	10	11	12
Number of orders	180	182	129	146	138	45	45	125	148	132	195	202

Q4(d) Worked Solution:

Work out both the median **and** the mode for Rani **and** give one reason why the median is the better average for her to use.

[4 marks]

45 45 125 129 132 138 146
or 202 195 182 180 148 146 138
142 (median)
45 (mode)
Any valid comment
[4 marks]

Your answer:

Past Paper

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4 (e) The company's income in 2019 was £863 490

This was 7% higher than in 2018

Work out the company's income in 2018

[2 marks]

Q4(e) ~~Worked Solution:~~
(£)807 000
863490 ÷ 1.07
[2 marks]
Tip: Award 2 marks if correct answer given

Your answer: **£**

Past Paper

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4 (f) The company also prints t-shirts.

$\frac{2}{11}$ of the t-shirts are white.

$\frac{1}{3}$ of the remaining t-shirts are black.

What is the ratio of white : black t-shirts?

[2 marks]

Q4(f) Worked Solution:

2 : 3
3
or 2 : 9
11
10
[2 marks]
Tip: Award 2 marks if correct answer given

Your answer:

[Total marks: 15]

This is the end of the assessment.

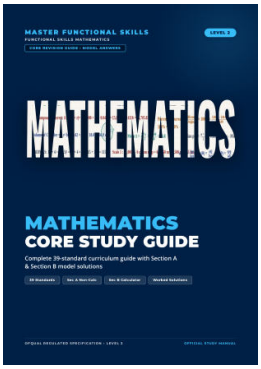
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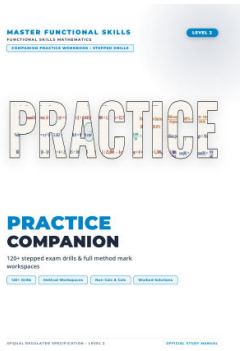


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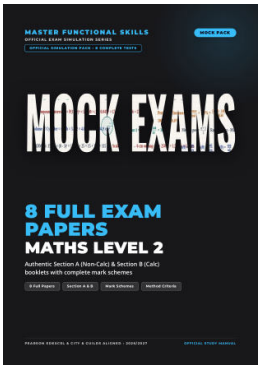


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